Aerial Robotics Coursework

EEEE4138 | Individual MATLAB and Simulink project

Build, validate and control your assigned four-rotor aircraft in MATLAB and Simulink. Compare cascaded PD control with a state-feedback design and explain the effects of payload, drag and motor response.

Assessment: individual coursework, marked out of 100 and worth 40% of the module. Submit by 20th December 2026, 11.59PM through Moodle.

Assessed learning outcome LO4: Use modern software tools (like Matlab/Simulink) for analysis and control design of aerial robots.

What you will submit

- One PDF report: EEEE4138CW_StudentID.pdf. Organise it by Tasks 1–3, followed by conclusions and references. About 15 pages of main text is a guide, not a page-limit penalty.
- One ZIP: EEEE4138CW_StudentID.zip. Include all models, parameter scripts, functions, reference generators, run scripts and a README. State the MATLAB release and how to reproduce the figures and results. Use relative paths.
- Put your assigned parameter ID on the report and in the parameter script. Check that the submitted package runs from a clean folder.

Aircraft specification

Use your allocated plus or X quadrotor and the m0 and l values from your parameter allocation. All aircraft have four rotors. m0 is unloaded mass including battery and equipment; l is centre-to-rotor distance. Keep parameters in one script.

Quantity	Required definition
Nominal and increased payload	Nominal payload = 0.50m0; total mass mb = 1.50m0. Increased payload = 0.80m0; total mass = 1.80m0.
Principal inertia	J = diag(0.030,0.030,0.050) [m0/(1.50 kg)] [l/(0.250 m)]^2 kg m². Treat payload as a point mass at the centre of mass; keep J unchanged in the payload test.
Rotor coefficients and limits	g = 9.81 m/s²; Omega_max = 800 rad/s; kT = 2mbg/(4Omega_max^2); kMT = 0.020 m; kM = kMT*kT. Ti = kT*Omega_i^2. Keep coefficients and limits fixed when payload changes.
Drag and motor lag	Still-air drag: Fdrag_body = -c_drag*vb, with c_drag = 0.25 N s/m. Motor lag: tau_m*dOmega_i/dt + Omega_i = Omega_cmd_i, with tau_m = 0.020 s. Enable drag and lag only in specified tests.

These are assigned simulation parameters, not measured aircraft or propeller data. Use the specified coefficients throughout.

Frames and rotor numbering

Use North–East–Down position and body axes x forward, y right, z down. Height h = −pD. Use ZYX yaw–pitch–roll angles and radians internally. Each rotor thrust acts along negative body z.

Rotor	Plus position	X position	Spin from above
1	(l,0,0) front	(d,−d,0) front left	CCW
2	(0,l,0) right	(d,d,0) front right	CW
3	(−l,0,0) rear	(−d,d,0) rear right	CCW
4	(0,−l,0) left	(−d,−d,0) rear left	CW

For X geometry, d = l/√2. Use reaction-torque signs [+1,−1,+1,−1] about body +z. The baseline excludes ground contact, wind, sensor noise and rotor gyroscopic moments. State these assumptions.

Task 1 Modelling and validation

1. Present your parameter table, one application of the configuration and one design trade-off. Explain assumptions and units.

2. Implement the nonlinear rigid-body model with four actual rotor speeds as inputs. Include position kinematics, body-velocity dynamics, attitude kinematics, rotational dynamics and gravity. State all states and initial conditions.

3. Derive the rotor thrust/reaction torques and force/moment allocation. Check its signs and independent thrust, roll, pitch and yaw control.

4. Compare analytical predictions with four isolated tests: zero-thrust free fall, equal-thrust hover, a differential-thrust moment, and a known frame transformation. Then test increased payload at the original hover rotor speeds. Separately, test drag with a nonzero initial horizontal velocity and nominal hover thrust. Explain discrepancies.

Task 2 Cascaded PD control and tracking

1. Implement outer position and inner attitude PD loops, hover feedforward, heading handling, rotor allocation and speed limits. Justify gains and relative loop speeds. Verify small commands before tracking paths.

2. Generate and track Cases A–C on the next page. Check reference derivatives and continuity, not just the aircraft response. Use ideal plant states for feedback.

3. Repeat Case A with increased payload only, then drag only. Keep nominal gains, mass compensation and rotor limits unchanged for the first comparisons.

4. Add a first-order lag with time constant 0.020 s to each rotor-speed command for a separate nominal-mass Case A test. Limit commanded speeds before the lag and initialise actual speeds at nominal hover. Test unchanged gains first. If retuning is needed, report an additional comparison; otherwise justify retaining the gains. Do not add another thrust lag.

The required baseline is PD. Integral action is optional; retain PD results and use anti-windup if adding integral control. Extra complexity alone earns no extra marks.

Task 3 State-space design and comparison

1. Linearise the 12-state rigid-body model at nominal, level stationary hover with zero yaw, no drag and no motor lag. State equilibrium, state order, inputs, outputs and A, B, C, D. Use perturbations in total thrust and body moments as the four inputs. Connect these to the nonlinear rotor-speed plant through allocation.

2. Evaluate controllability and observability of this 12-state model using outputs [pN,pE,pD,psi]. Explain the ranks and numerical tolerance. Ideal plant states are available for feedback; an observer is not required.

3. Design state feedback using a taught method. Analytical pole placement for cascaded channels is acceptable. State channel models and gains, verify local stability and explain the complete controller. Restore equilibrium/feedforward terms and account for changing heading. Explain any equivalence to PD: a justified equivalent design is acceptable, and better performance is not assumed.

4. Compare with PD for nominal Cases A–C and separate Case A payload and drag tests. Use identical plant, references, limits and numerical settings. Explain performance and local-model limitations.

Reference trajectories and required evidence

Initialise position at the start of each path, with zero velocity, zero roll/pitch and zero body rates. Use initial yaw pi rad for Cases A and B, facing the centre, and zero yaw for Case C. Initialise motors at nominal hover speed. The launch plane is a coordinate origin; no ground contact is modelled. Hold the final position and heading reference for 10 s.

$$S(s) = 10s^3 - 15s^4 + 6s^5, \quad s = (t - t0)/D, \quad 0 \le s \le 1$$

For a straight rest-to-rest segment from a to b, use rd = a + (b−a)S(s). Hold endpoints outside the interval. Derive velocity and acceleration analytically. This is scalar minimum-jerk timing; its circular mapping is not a claim of globally minimum Cartesian jerk.

Case A Circle

Start (N,E,h) = (25,0,0). Climb to h = 100 m over 60 s. Complete one circle of radius 25 m centred on (0,0), at h = 100 m, over 120 s: α = 2πS(s), N = 25cosα, E = 25sinα. Face the centre.

Case B Helix and circle

Start (N,E,h) = (20,0,0), in metres. Follow a radius-20 m helix. Ramp upward speed from zero to vc = 1/6 m/s over 30 s using S; hold vc for 570 s; ramp to zero over 30 s using 1-S. Integrate speed to obtain height: 100 m at 630 s. Use alpha = 2*pi*h/(20 m), giving five turns.

Transfer radially from (20,0,100) to (25,0,100) over 30 s using S, then perform the 120 s Case A circle. Face the centre throughout.

Case C Square

Start (0,0,0). Climb to 100 m over 60 s. Follow (N,E) vertices (0,0), (50,0), (50,50), (0,50), (0,0), taking 30 s per side using S. Velocity and acceleration are zero at every corner. Keep yaw zero.

For A and B, inward yaw is atan2(-E,-N). Maintain a continuous unwrapped reference beginning at pi rad, and wrap the yaw feedback error to [-pi,pi]. Do not differentiate wrapped angle jumps. Heading rate is zero during vertical-only and radial segments.

Evidence for each required tracking run

For each run, plot reference and actual position and yaw, actual roll/pitch and rotor speeds. Include desired roll/pitch if generated. Report position RMSE = sqrt(mean(||p-pd||^2)), maximum position-error magnitude, wrapped yaw RMSE, peak actual rotor speed, and the percentage of uniformly logged samples with any limited rotor-speed command at a limit. State units and the limit-detection tolerance.

Use identical sampling grids and comparison intervals for matched runs. Compare actual and reference values at the same logged times. Include the final 10-second hold in the overall metrics. For Case B, also report the metrics separately for the helix climb, radial transfer, circle and final hold. Calculate the limit statistic from rotor-speed commands after clipping and before motor lag. Explain persistent tracking errors, saturation and unsuccessful runs.

Record solver settings and tolerances. For motor lag, begin with maximum step 0.005 s and halve it on one demanding case to check convergence. Task 2 requires six baseline runs: nominal A, B, C and separate Case A payload, drag and motor-lag tests. Add a run if retuning. Task 3 requires five: nominal A, B, C and separate Case A payload and drag tests, with motor lag disabled. Reset initial conditions for each run.

Assessment rubric

The four criteria total 100 marks and contribute 40% of the module mark. Assess each criterion using evidence presented in the report. Choose a whole-number mark within the band that best fits the evidence. Award zero where there is no relevant assessable evidence.

Criterion	Poor	Satisfactory	Good	Excellent
Task 1 Modelling and validation 30 marks	0–11 Missing or incorrect equations, allocation or signs undermine the reported validation. Few usable results.	12–17 Shows a basic model and some checks. Predicted values, simulated values or explanations are missing for several tests.	18–23 Shows a mostly correct model and allocation. Compares predictions with simulated values for most required checks and explains the main differences.	24–30 Derives the model and allocation correctly. Compares predictions with numerical results across the required checks and explains agreement or discrepancies using signs, units and assumptions.
Task 2 PD control and tracking 35 marks	0–13 Controller or references contain major errors, or the report gives little usable evidence of tracking.	14–20 Shows basic tracking. Gain justification, reference checks or required comparisons have substantial gaps.	21–27 Reports most required runs with relevant metrics. Explains gain choices and the main effects of payload, drag or motor lag.	28–35 Justifies gains and verifies reference derivatives. Uses matched runs, metrics and logged signals to explain payload, drag and lag effects, including any trade-offs or limitations.
Task 3 State-space design 25 marks	0–9 Linearisation or feedback design contains major errors. Rank or performance claims have little supporting evidence.	10–14 Presents a basic linear model and feedback design. Rank interpretation, stability evidence or comparisons are incomplete.	15–19 Gives a mostly correct hover model, interprets the ranks and supports the feedback design. Compares performance under mostly matched settings.	20–25 Derives the hover model correctly and explains rank tolerance and stability. Interprets matched comparisons using metrics and states local-model limits. A justified PD-equivalent design can earn full marks.
Report and reproducibility 10 marks	0–3 Methods or results are hard to locate. Figures or instructions give too little evidence to trace the reported results.	4–5 Report is readable, but labels, settings or instructions leave gaps in how results were obtained.	6–7 Report links most conclusions to labelled evidence. Documents the MATLAB release, settings and how to reproduce the main results.	8–10 Report links claims to labelled figures or tables. Gives clear run steps, settings and file references for reproducing the reported results.